

Varun Saini

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PROFESSIONAL SUMMARY

Data and analytics engineer with experience in Python, Data Analysis, Machine Learning. Experienced in building ETL pipelines, model validation and engineering-focused data analysis for operational insights.

TECHNICAL SKILLS & TOOLS

Data & Analytics: Python, Pandas, Machine Learning, ETL

Engineering Automation: Automation, Streamlit, Model Validation

Documentation: Technical Documentation

WORK HISTORY

LOGISTICS SUPERVISOR

Booths | United Kingdom

- Built ETL processes and data cleaning pipelines to support inventory analytics and reporting.
- Developed automated dashboards and scripts to reduce manual reconciliation effort.
- Provided structured datasets for machine learning and predictive analytics projects.

RESEARCH INTERN

University of Lancashire | Preston, United Kingdom

- Implemented data preprocessing and MATLAB scripts to clean simulation outputs for analysis.
- Performed data-driven validation and contributed to model development workflows.
- Developed reproducible analysis pipelines to support engineering research.

VOLUNTEER CITIZEN SCIENTIST

Zooniverse

- Reviewed and labelled large astronomy datasets to produce training data for classification models.
- Contributed to data quality checks and supported downstream analysis workflows.
- Documented classification criteria to improve consistency across contributors.

VOLUNTEER CITIZEN SCIENTIST

NASA Globe Mission

- Contributed to environmental and climate observation workflows through scientific data collection and analysis.

CAD DESIGNER (2D & 3D)

Kaizen Industries | Remote (Part-Time)

- Delivered detailed SolidWorks 2D and 3D CAD models for manufacturing-ready mechanical assemblies.
- Produced technical drawings, assembly documentation and build-ready fabrication specifications.
- Coordinated design revisions and review cycles with engineering and production teams.

CLASS REPRESENTATIVE

University of Lancashire | Preston, United Kingdom

- Represented student engineering cohorts, coordinated academic feedback and organised technical activities.

PROJECTS

Warehouse Intelligence & Inventory Analytics Platform

Implemented ETL and analytics pipelines to process 243000 warehouse movement records for predictive inventory analytics.

Python, Pandas, Streamlit, Plotly, GitHub

Aircraft Predictive Maintenance AI System

Built ML pipelines and feature engineering workflows for predictive maintenance using NASA CMAPSS datasets.

Python, NASA CMAPSS Dataset, Machine Learning

AI Life Coach & Sentiment Analysis System

Developed NLP pipelines and sentiment analysis models for personalised coaching recommendations.

Python, NLP, Sentiment Analysis

EDUCATION

BEng with Honours in Aerospace Engineering with Pilot Studies (Graduated 2026)

CAD, Structure & FEA, Propulsion, Pilot Studies, Thermo-Fluids, Aeromechanics, Aircraft Design, Project Management

University of Lancashire, United Kingdom

Diploma of Higher Education: Mechanical Engineering (09/2022 – 07/2023)

Electrical and electronic principles, Engineering Mathematics, Strength of Materials, Mechanical Design and Engineering Applications, Mechanical Science, English Literature

PSB Academy, Singapore

German Language Course (08/2020 – 08/2022)

Language Pantheon, Delhi

Certificate of Higher Education: Medical and Engineering (05/2018 – 06/2020)

Himalaya Public Sr. Sec. School, Delhi

CERTIFICATIONS

- NASA: Learning from Dark Matter and Energy
- NASA: Explore Space Tech: Rockets 101
- NASA: Technology Drives Exploration: Detecting Exoplanets
- IARC: Life in Space

PROFESSIONAL MEMBERSHIPS

IMechE, The Institute of Engineers Singapore, NVIDIA Developer Program, JAXA ALOS Global User

LANGUAGES

English (Fluent), Hindi (Native), German (Intermediate)

References available on request